Measure-to-measure interpolation using Transformers: formalization

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0.1 Foundations

Definition 1 (Tangential projector)**.** The layer-normalization projector $P_x^\perp v = v - \langle x, v\rangle x$ onto the tangent space.

Lemma 2 (Projector identity, L1)**.** *For* $\|x\| = 1$, $P_x^\perp$ *is symmetric and idempotent,* $P_x^\perp x = 0$*, and* $\langle P_x^\perp v, v\rangle = \|v\|^2 - \langle x, v\rangle^2$*. [machine-checked]*

Definition 3 (Geodesic distance)**.** $d_g(x, y) = \arccos\langle x, y\rangle$, with $\cos d_g(x, y) = \langle x, y\rangle$ for unit vectors. [machine-checked]

0.2 Labeled axioms (missing Mathlib infrastructure)

Definition 4 (Wasserstein distance)**.** W_2, W_1 with nonnegativity, symmetry, triangle inequality, the map-coupling bound, and Kantorovich-Rubinstein duality. [axiomatised]

Definition 5 (Continuity-equation flow)**.** The solution map Φ_θ^t on points and on measures, with Lipschitz, bijective and parked-region-identity properties; the switch count. [axiomatised]

0.3 Self-contained leaves

Lemma 6 (Gate algebra and gate ODE, L2)**.** $\frac{d}{dt}\langle x(t), \omega\rangle = (\cos R - \langle z, x\rangle)_+ (1 - \langle x, \omega\rangle^2)$ *along the gated flow, with the gate active exactly on* $\{d_g(z, x) > R\}$ *(eq. B.4-B.5). [machine-checked] The kernel-checked equivalence* $g(x) > 0 \iff d_g(z, x) > R$ *corrects the printed (B.4), which states the gate is active on* $\mathcal{B}_0 = B(z, R)$*; consistency of Lemma B.2 requires the opposite sign (finding F1 in* `RESEARCH.md`*, confirmed independently by experiment E1).*

Lemma 7 (Barycenter ODE, L6)**.** $\frac{d}{dt}\langle x, \alpha\rangle = c\,(1 - \langle x, \alpha\rangle^2)$ *and strict increase for* $c > 0$ *(eq. B.9). [machine-checked]*

Lemma 8 (Geodesic gradient, L4)**.** $\frac{d}{dt}d_g(x(t), \omega) = -(1-\langle x, \omega\rangle^2)^{-1/2}\langle x', \omega\rangle$*, exhibiting* $-P_x^\perp\omega/\sqrt{1 - \langle x, \omega\rangle^2}$ *as the Riemannian gradient (4.4). [machine-checked]*

Lemma 9 (Separating hyperplane, L3)**.** $d_g(\omega, x) \geq \pi/2$ *and* $\tau \in (0, 3\pi/8)$ *give* $\langle \omega, x\rangle < \cos(\pi/8+\tau)$ *(Prop 4.2 Step 1). [machine-checked]*

Lemma 10 (Lyapunov sign, L5)**.** $E = 1 - \cos\theta$ *along* $\theta' = -\alpha\sin\theta$ *has* $\dot{E} = -\alpha\sin^2\theta \leq 0$ *(Ex. 6.1). [machine-checked]*

Lemma 11 (Ball-chain retention, L9)**.** *Per-step retention* $(1-\varepsilon)$ *gives* $a_K \geq (1-\varepsilon)^K a_0$ *(Lemma B.1 induction). [machine-checked]*

Lemma 12 (Pigeonhole, L10)**.** *A nonempty open ball contains a point* $\neq a$*, so no map is constant on it (Lemma 3.4 Part 1). [machine-checked]*

Lemma 13 (Disjoint hulls give non-colinear barycenters, L11)**.** *Modeling* $\mathrm{conv}_g(s) = \mathrm{cone}(s) \cap \mathbb{S}^{d-1}$ *(faithful in a hemisphere), empirical measures in the orthant with disjoint geodesic hulls have non-colinear barycenters (not* SameRay*). Discharges the step Proposition 3.1's induction asserts without proof (finding F2); kernel-clean for the empirical regime of Thm 1.1 / restricted Thm 1.2. [machine-checked]*

Lemma 14 (Linearized OT bound, L7)**.** $W_2(T^1_\#\mu, T^2_\#\mu) \leq \|T^1 - T^2\|_{L^2(\mu)}$ *(Lemma 5.2). [axiomatised]*

Lemma 15 (Markov bound, L8)**.** $W_2(\mu, \delta_{x_0}) \leq \eta_2 \Rightarrow 1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$ *(Claim 2). [informal]*

0.4 Mass transport and clustering

Lemma 16 (Single ball pair, B.2)**.** *Mass* $\mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1-\varepsilon)\mu_0(\mathcal{B}_0)$*. [informal] Caveat (F1): the printed parameters* $U = -z\mathbf{1}^\top,\ b = \cos(R)\mathbf{1}$ *make the gate active on the complement of* $\mathcal{B}_0$*, so the lemma transports nothing as written; the sign must be flipped to* $U = +z\mathbf{1}^\top,\ b = -\cos(R)\mathbf{1}$*. True once corrected.*

Lemma 17 (Ball chain, B.1)**.** *Mass through* K *overlapping balls* $\geq (1-\varepsilon)^K$*. [informal]*

Proposition 18 (Clustering to a point, 2.1)**.** *A single-hemisphere measure clusters to a point;* $\mathrm{diam} \to 0$ *at rate* $O(\log 1/\varepsilon)$*. [informal]*

Proposition 19 (Clustering to discrete, 2.2)**.** *Disjoint-support atomless measures cluster to* M *atoms. [informal]*

0.5 Disentangling supports

Lemma 20 (Transport to the orthant, 3.2)**.** *One switch moves measures into* $\mathbb{Q}_1^{d-1}$*. [informal]*

Lemma 21 (Disentangle non-colinear, 3.3)**.** *Shrink a measure's hull to its barycenter direction. [informal]*

Lemma 22 (Make non-colinear, 3.4)**.** *Two measures made non-colinear. [informal]*

Proposition 23 (Disentanglement, 3.1)**.** *A piecewise-constant* θ *renders the* N *supports pairwise disjoint,* $O(dN)$ *switches. [open] The "disjoint hulls* $\Rightarrow$ *non-colinear barycenters" step the induction asserted without proof (finding F2) is now the machine-checked leaf L11 (13) for the empirical regime; the headline stays open only because it rests on the flow /* conv_g*-nesting axioms.*

0.6 Matching discrete measures

Proposition 24 (Steer one point, 4.2)**.** $x_0^M \to y^M$ *while inactive points stay fixed,* ≤ 6 *switches. [informal]*

Proposition 25 (Match ensembles, 4.1)**.** M *points* $x_0^i \to y^i$ *exactly,* $\leq 6M$ *switches. [informal]*

0.7 Main results

Lemma 26 (Transport maps preserved, 5.1)**.** *Existence of* ψ*. [informal]*

Lemma 27 (L^2 approximation, 5.4)**.** *Approximate any* ψ *by a flow map. [informal]*

Theorem 28 (Theorem 1.2)**.** *A single Transformer matches* N *arbitrary inputs to* N *arbitrary (transport-matchable) targets in* W_2*. [open]*

Theorem 29 (Theorem 1.1)**.** *Dirac targets: matching with* $O(dN)$ *switches and norm* $O(dN/T + \log 1/\varepsilon)$*. [open]*

Proposition 30 (Generic O(1) switches, 6.2)**.** *For generic i.i.d. discrete inputs a single constant parameter disentangles almost surely. [informal]*